MILITARY SPECIFICATION

WATER BATH, ELECTRIC, Serological, 6 Rack, 115 volt, 50-60 Hz, AC

This limited coordination Military specification has been prepared by the Defense Personnel Support Center based upon currently available technical information, but it has not been approved for promulgation as a coordinated document. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers an electrically heated, thermostatically controlled water bath, suitable for use in serological testing (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-N-281	Nickel-copper-alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings and Structural and Special Shaped Sections
GG-T-336	Thermometer, Self-Indicating, Liquid-in-Glass (Laboratory)
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Shipping, Fiberboard.

Military

MIL-M-7298	Manual, Technical: Commercial Equipment.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

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STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C73.11	Dimensions of Caps, Plugs and Receptacles, 125 volts, 12 Amperes, 2 Pole, 3 Wire, Grounding Type, General Purpose.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

UNDERWRITERS' LABORATORIES, INC. (UL)

499	Standard for Heating Appliances, Electric.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, L.I., NY 11749; 207 E. Ohio Street, Chicago, IL 60611; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70	National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

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3. REQUIREMENTS

3.1 Fire and casualty hazards. The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of UL 499. Proof shall consist of one of the following:

a. UL label or UL listing of the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the contracting officer, upon the latter's request.

b. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of test. Detailed findings of examination necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.1.2 Electrical parts. All electrical parts and wiring shall conform to the requirements of the National Electrical Code and the applicable standards of the Underwriters' Laboratories, Inc.

3.2 Materials. Materials shall conform to the applicable specifications and requirements as specified herein. Materials not specifically designated shall be of high commercial quality throughout, and shall be entirely suitable for the purpose intended.

3.2.1 Corrosion-resisting steel. Chemical composition of corrosion-resisting steel shall comply with FED-STD-66, AISI 300 series; and class 302 or 304 for the test tube racks (see 3.3.2).

3.2.2 Nickel-copper alloy. Chemical composition of nickel-copper alloy (monel) shall comply with QQ-N-281, class A.

3.2.3 Insulation. Insulation shall be non-hygroscopic, non-powdering and non-packing material having a thermal conductivity coefficient not exceeding 0.27 BTU-inch per hr.-sq. ft. -^{o}F.

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3.3 Construction. Water bath shall be double-walled with insulation between walls. Exterior walls shall be of corrosion-resisting steel 0.025 inch thick No. 24 United States Standard (USS) gage. Exterior dimensions shall be 15 ± 1/4 inches (in.) long, 14 ± 1/4 in. wide and 6-1/2 ± 1/4 in. high. Nominal interior (water chamber) dimensions shall be 13 in. long, 12 in. wide and 4-5/8 in. deep. Design shall be such that the water chamber accommodates 6 Wassermann type test tube racks conforming to 3.3.2. For purposes of this specification, the portion which holds the test tube racks will be defined as water chamber, and the entire unit as water bath. Water chamber shall be fabricated of monel .018 in. thick 26 USS gage. The water chamber and outside body shall be of spotwelded construction. Water chamber shall be soldered watertight. The water bath shall have a removable base fabricated of aluminum .063 in. thick 14 Brown and Sharpe gage. The unit shall be capable of operating on 115 volts (nominal), 50/60 cycle AC, 400 watts. Water bath shall consist of the parts specified in 3.3.1 through 3.3.8 and any other parts required for specified performance.

3.3.1 Cover. Cover shall be fabricated of corrosion-resisting steel .025 in. thick 24 USS gage and shall be the gable type. Cover shall have a plastic handle on the top and a cutout on one side. Cutout shall provide space for thermometer holder and thermometer. In order to prevent the cover from sliding off or into the water chamber, the bottom of cover shall protrude into the water chamber and shall have a flange on sides which rest on the top of water bath. The design and dimensions shall be such that the cover can be easily set in place and can be easily removed.

3.3.2 Test tube racks. Each unit shall be supplied with 6 removable test tube racks constructed from a minimum .018 in. thick USS gage corrosion-resisting steel; commercial tolerances shall apply. The test tube racks shall conform to Figure 1 for design and dimensions. The free edges of the shelves and end walls shall be flanged. The shelves shall also be flanged at the ends, fitted inside the flanged end walls and affixed by not less than two spot welds at each end of each shelf. Holes shall be numbered in pairs from 1 to 10 lengthwise along the center of the top shelf between the 2 rows. The test tube racks shall be free of sharp edges, pits, burrs, dents and distortion.

3.3.3 Thermometer holder. Thermometer holder shall be fabricated of corrosion-resisting steel .038 in. thick 20 USS gage. Shall have hole with inserted rubber ring to hold the thermometer safely.

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3.3.4 Thermometer. Each unit shall be provided with a thermometer in accordance with GG-T-336, Type II. Thermometer shall be graduated in 1° Centigrade (°C.) increments from 0° to 100°C. Shall have the following nominal dimensions: 1/4 in. in diameter, 10 in. long. The thermometer readings shall be accurate within ± 0.5°C.

3.3.5 Control panel housing. Control housing shall be plastic and shall be located on front side of water bath. Shall have a dial plate, thermostat control knob, switch and pilot light.

3.3.5.1 Dial plate. Dial plate shall have reference markings on a circular graduation scale for thermostat setting. Every fifth graduation shall be numbered in the following manner: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 0.

3.3.5.2. Thermostat control knob. Shall be a plastic knob having an indicating mark on the lip. Various thermostat settings shall be obtained by turning the knob.

3.3.5.3 Switch. Switch shall be "on-off" line type, single pole, 15 amperes, 125 volt, AC.

3.3.5.4 Pilot light. Pilot light shall light when heating elements are energized and shall not light when heating elements are not energized.

3.3.6 Heating element. Two strip heating elements shall be located in close contact with exterior surface of the bottom of the water chamber. Each heating element shall be rated 120 volt, 200 watts.

3.3.7 Thermostat. Thermostat shall be adjustable hydraulic type with copper bulb and capillary. Bulb shall be held in close contact with the heater and the exterior surface of the bottom of the water chamber.

3.3.8 Electrical cord. Each unit shall be provided with an electric cord no less than 6 feet in exposed length. The cord shall be 3-wire with a 3-prong plug on one end, suitable for use with receptacles specified in ANSI C73.11.

3.4 Performance.

3.4.1 Temperature accuracy. The indicated temperature shall be constant ± 0.5°C. (highest temperature shall not exceed lowest temperature by more than 1°C.), when tested as specified in 4.4.1.

3.4.2 Heating time. The water bath shall reach a constant temperature within 2-1/4 hours, when tested as specified in 4.4.2. (See 3.4.1 for definition of constant temperature.).

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3.4.3 Dielectric strength. The water bath shall have no electrical breakdown, when tested as specified in 4.4.3.

3.4.4 Range. Water in uncovered water bath shall reach a temperature of at least 60°C. and water in the covered water bath shall reach the temperature of boiling water, when tested as specified in 4.4.4.

3.5 Service data. The contractor shall furnish with each unit two copies each of a technical manual which contains complete instructions for installation, operation, and maintenance and lists of component parts. The manual shall comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed technical manual, for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary manuals shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract and propose to furnish manuals identical with those previously accepted, the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of 15 additional copies shall be at the option of the Government.

3.6 Identification marking. Each bath shall have a nameplate on which shall be permanently and legibly marked the following information: manufacturer's name or trademark, contract or purchase order number, National Stock Number, "U.S.", the voltage, wattage and serial number. Markings secured by adhesive shall not be used.

3.7 Workmanship. Workmanship shall be first class throughout. Soldering and welding shall be free from cracks and porosity. The bath shall be free from sharp edges, pits, burrs, cracks, dents and other defects which may detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection

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requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the steel, nickel-copper alloy or thermometer, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL) shall be as indicated in Table I. Unit of product shall be the water bath.

Table I Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

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4.2.2 Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105 and Table II. The unit of product shall be one complete water bath.

Table II Sampling for tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Temperature accuracy	3.4.1	4.4.1	S-2	*
Heating time	3.4.2	4.4.2	S-2	*
Dielectric strength	3.4.3	4.4.3	100%	**
Range	3.4.4	4.4.4	S-1	*

* Acceptance number shall be zero.
** Failure of one unit shall be cause for rejection of the lot.

4.3 Examination. The unit shall be examined to determine compliance with all requirements contained in the specification.

4.3.1 Classification of defects. The water bath shall be examined for defects including, but not limited to, those listed in Table III.

Table III Classification of defects

Categories	Defects
Major	
101	Bath not completely assembled or cannot be properly assembled with all component parts.
102	Bath (components) not free of cracks, tears, dents, deformation or corrosion.
103	Bath not free of sharp edges.
104	Bath not free of wobble and rocking when placed on a flat and level surface.
105	Applied soldering or welding not free of cracks and/or porosity.
106	Line cord not adequately secured to housing for water bath.
107	Line cord not free of cuts, tears or abrasion.
108	Blades of line cord plug not properly aligned.
109	Thermostat settings cannot be adjusted throughout entire range.

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Classification of defects (cont'd)

Categories	Defects
Minor	
201	Identification marking not complete, legible or permanent.
202	Bath (components) not free of dirt, grease or foreign matter.
203	Surfaces of water bath not free of scratches which detract from its appearance.

4.3.2 Dimensional examination. The water bath shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Temperature accuracy. The thermostat shall be set so that the water in the bath attains the following temperature: 37 ± 2°C. and 56 ± 2°C. Cover shall be in place and amount of water and location of sensing portion of the temperature-detecting device shall be in accordance with service data submitted by the contractor. The following procedure shall be followed for 37°C. and 56°C. settings: After a stable temperature has been reached, temperature readings shall be recorded continuously until the pilot light has been energized 5 times. Maximum time interval between recorded readings shall be 10 seconds. Temperature-detecting device shall have response time of 3 seconds (maximum) and shall indicate temperature changes of 0.1°C. (see 6.1). Response time shall be defined as the interval of time required for temperature-detecting device to adjust to and indicate a change of 0.1°C.

4.4.2 Heating time. The water bath shall be filled with water 20 ± 1°C., and the cover shall be put in place. The thermostat shall be set so that bath will attain 56 ± 2°C. and heating system shall be turned on. The time for the thermometer to indicate the selected temperature shall be monitored and recorded.

4.4.3 Dielectric strength. While the water bath is in a heated condition, twice the rated voltage plus 1000 volts (a total of 1240 volts) shall be applied between the current-carrying parts and the metal enclosure. The applied potential shall be increased gradually until the required value is reached, and shall be maintained at the required value for 1 minute.

4.4.4 Range. Water and thermometer shall be placed in uncovered water bath in accordance with service data submitted by the contractor. With the cover off, the thermostat shall be set to its highest graduation, the heating system shall be energized and allowed to heat until a constant temperature is reached (see 3.4.1). Any water lost by evaporation shall be replaced and cover shall be put in place. Water bath shall be allowed to continue to heat until a constant temperature is reached (see 3.4.1). Ambient temperature shall be 20 ± 1°C.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One water, bath, as specified, constitutes one unit of issue.

5.1.2 Level A. Packaging shall be Level A or C as specified (see 6.2).

5.1.2.1 Unit preparation. The thermometer shall be packaged in a plastic or spiral-wound chipboard container of suitable size. Container design shall include suitable interior fittings or shall contain sufficient cushioning material to insure against damage to the contents. The thermometer along with the test tube racks shall be secured within the water bath chamber in such a manner as to prevent shifting of, or damage to, the contents or unit. The water bath shall be wrapped in kraft paper and cushioned with corrugated fiberboard or other suitable material to insure against marring, scratching or damage to the unit.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Each unit, packed in an exterior container, shall be designed for a type 2 load, and constructed in accordance with PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproofed as specified in 30.4 of PPP-B-636.

5.2.2 Level B.

5.2.2.1 Exterior container. Each unit shall be packed in an exterior container conforming to PPP-B-601, domestic type; or shall be designed for a type 2 load and constructed in accordance with PPP-B-621, class 1; or PPP-B-636, type CF,

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class domestic. Closure shall be in accordance with appendix of applicable box specification. Special requirements of applicable table shall apply for PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 A temperature-detecting device which meets the response time and sensitivity requirements of 4.4.1 is Hewlett Packard Model 2801-A, Quartz Thermometer.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification covers the following item as listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-765-0621	WATER BATH, ELECTRIC, Serological, 6 Rack, 115 volt, 50-60 Hz, AC

6.4 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types normally procured to meet requirements.

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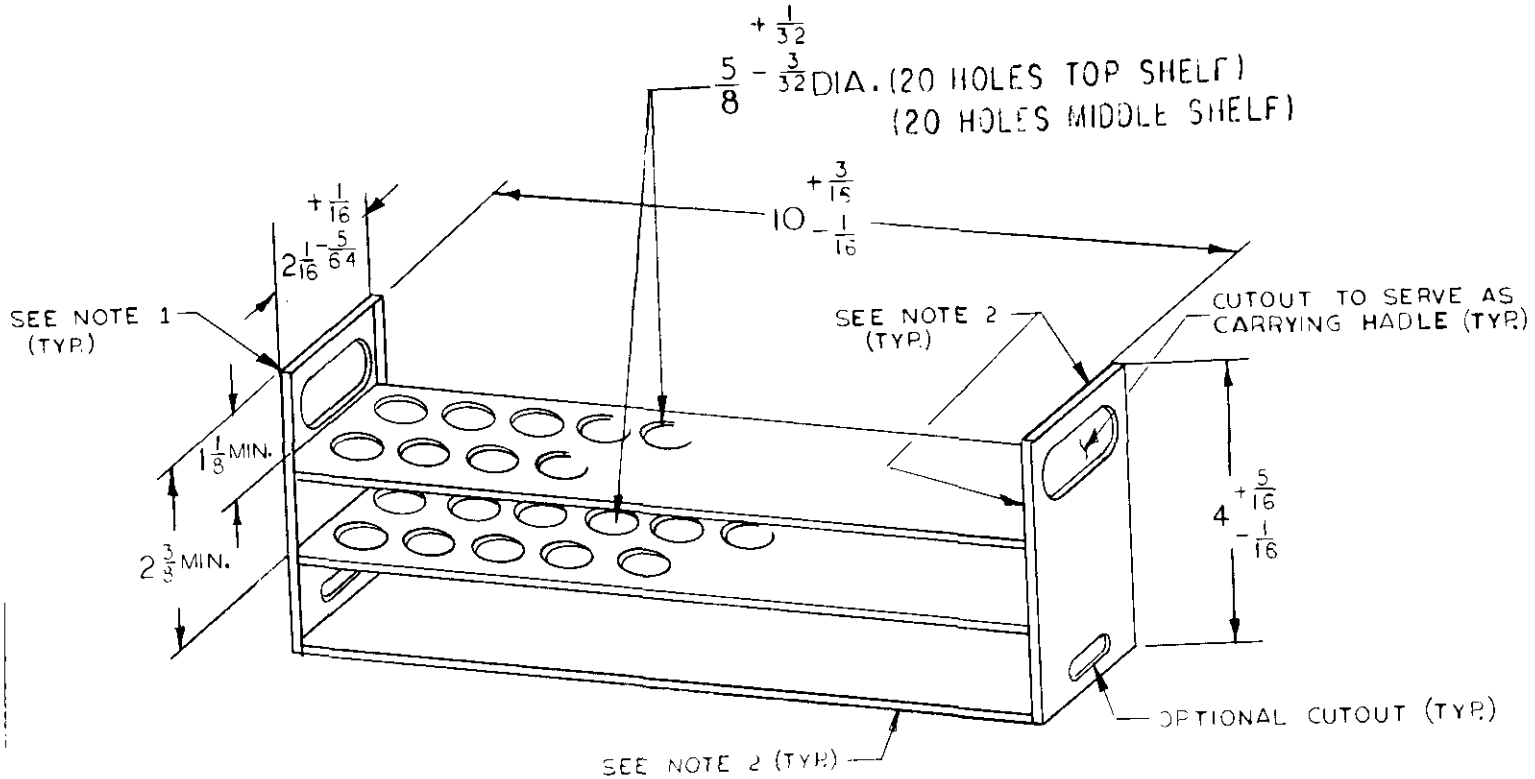
Review activities:
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TEST TUBE RACK

FIGURE 1

NOTES

1. AS AN ALTERNATE TO THE CLOSED FLANGE CORNERS, OPEN CORNERS WITH SMOOTH EDGES SHALL BE ACCEPTABLE.
2. EXCEPT AS INDICATED IN NOTE 1, FREE EDGES OF SHELVES AND END WALLS SHALL BE FLANGED A MINIMUM OF 1/4 INCH.
3. ALL INDICATED DIMENSIONS ARE IN INCHES.

TEST TUBE RACK

FIGURE 1

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DOCUMENT IDENTIFIER AND TITLE
MIL-W-37045 (DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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